



Long-term PM_{2.5} exposure in Bangladesh: identification of pollution hotspots, trends, sources and health risk assessment

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Abstract

Fine particulate matter (PM_{2.5}) represents a significant global challenge due to its severe effects on human health, climate, and the environment, and is identified as the leading cause of premature mortality and morbidity worldwide. The limited availability and distribution of ground-based measurements hinder long-term studies on the impacts of air pollution in Bangladesh. Therefore, in this study, global estimates of monthly PM_{2.5} developed by Washington University were used to analyze the spatiotemporal distribution and variability of PM_{2.5}, trends, health risk (HR), control zones, and potential source contribution function (PSCF) in Bangladesh from 2001 to 2020. The 20-year average spatial distribution shows PM_{2.5} hotspots across Bangladesh, particularly in the urban areas of Dhaka, Mymensingh, Chittagong, Barisal, Khulna, Rajshahi, and Rangpur, with higher pollution in the winter than in other seasons. In winter, PM_{2.5} ranged from 55.12 to 159.42 µg/m³ across 64 cities in Bangladesh, which is 11 to 32 times higher than the World Health Organization Air Quality Standards (WHOQS; annual mean: ≤ 5 µg/m³) and 4 to 11 times higher than the Bangladesh National Ambient Air Quality Standards (BNAAQS; annual mean: ≤ 15 µg/m³). Moreover, PM_{2.5} levels significantly increased in 63 cities across Bangladesh, with rates ranging from 0.54 to 1.38 µg/m³/year between 2000 and 2020. PM_{2.5} components in Bangladesh show an increasing trend for Sea Salt (SS), Organic Carbon (OC), Sulfate (SO₄), Black Carbon (BC), and Nitrate, except for Dust, which exhibits a negligible decreasing trend. Weather conditions, industrial emissions, vehicle exhaust, and biomass burning significantly influence PM_{2.5} concentrations. The HR assessment showed that the percentage of extremely high-risk areas in Bangladesh rose significantly from 14.57 to 39.29% between 2001 and 2020. Finally, PSCF analysis shows that air quality in Bangladesh is mainly affected by external sources of PM_{2.5} originating from the Indo-Gangetic Plain (IGB), India, Nepal, Bay of Bengal, Sri Lanka, Gulf of Mannar, Arabian Sea, and Laccadive Sea, with the strongest impact in the winter, followed by spring, autumn, and summer. It is recommended that policymakers utilize the findings of this study to implement targeted strategies for reducing PM_{2.5} concentrations and improving air quality across Bangladesh.

Keywords PM_{2.5} · Hotspot · Variability · Trends · Health impact · Bangladesh

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Introduction

Air pollution, specifically the increased levels of fine particulate matter (PM) with a dry radius smaller than $2.5\ \mu\text{m}$ ($\text{PM}_{2.5}$), presents a major concern studied in different regions across the World (e.g., Xu et al. 2020; Sarwar et al. 2023; Hagler et al. 2022). Highly populated Asian cities such as Dhaka in Bangladesh are notably susceptible to air pollution and its effects (Islam et al. 2023; Xing et al. 2016; Hopke et al. 2016; Kim et al. 2006). Bangladesh ranks ninth among seventeen countries with high outdoor ambient particulate matter levels, fourth in ozone exposure, and eleventh in household air pollution exposure (Faisal et al. 2022). Faisal et al. mentioned that, in recent years, Bangladesh has been coping with significant challenges posed by air pollution, particularly concerning $\text{PM}_{2.5}$. In general, the aerosol particles contributing to $\text{PM}_{2.5}$ are so small that they can penetrate deeply into the respiratory system and reach the lungs, pose serious health risks to humans such as eye, nose, throat and lung irritation, coughing, sneezing, running nose, and difficult breathing (Pope et al. 2002; Xing et al. 2016; Zhang et al. 2018; Shi et al. 2015; Burnet et al. 2018). These fine particles are a concern for health-related issues not only in Bangladesh, but all over the world, e.g., in Nigeria (Tariq et al. 2023a, b), in East and South Asia (Chen et al. 2019), in India (Goel et al. 2021), in China (Chen et al. 2023), in Japan (Yorifuji et al. 2006), in South Korea (Byun et al. 2024), in the USA (Kelly et al. 2019), and Europe (Thunis et al. 2023). These fine particles (i.e., $\text{PM}_{2.5}$), originating from both natural and human sources, are significant air pollutants contributing to deteriorating air quality in urban areas like the megacities of Dhaka and Delhi (Kumar et al. 2020). Anthropogenic sources such as transportation, industrial processes, power generation, commercial and residential energy consumption (including heating and cooking), diesel generators, construction activities, and road dust emissions directly contribute to the overall ambient levels of $\text{PM}_{2.5}$ (Kumar et al. 2017) and indirectly through gas-to-particle reactions (Hama et al. 2020).

The rapid and unplanned urbanization and industrial growth in Bangladesh have exacerbated the issue, leading to deteriorating air quality in many urban centers across the country (Rahaman et al. 2023). This process has brought significant challenges to public health as the urban environment has become a breeding ground for various health risks. Due to the absence of alternatives to bricks in construction activities, the number of brick kilns in Bangladesh has been steadily increasing, with an estimated annual growth rate of 5–6%, because of the upsurge in construction activities in recent years (DOE 2019). Dhaka, the capital city of Bangladesh and the second most populous city in the world, has experienced unprecedented urbanization in recent decades,

leading to numerous health risks and environmental challenges (Begum and Hopke 2018; Ribeiro 2021; Hama et al. 2020). In this city, vehicle emissions, surface dust, and brick kilns are the main sources of air pollution, amounting to about 85%, while artificial sources are more distributed and contribute significantly (Williams 2020). $\text{PM}_{2.5}$ emissions from brick kilns (17556.70 tons per year) are the top among other sources in Dhaka city (Ali et al. 2022). Vehicle emissions significantly contribute to air pollution in Bangladesh, particularly in Dhaka, where approximately 40% of the country's imported vehicles are registered (DOE 2019). In this city, $\text{PM}_{2.5}$ emissions from mobile sources (1404.60 tons per year) are the second highest (Ali et al. 2022). Overall, this capital city often experiences severe $\text{PM}_{2.5}$ levels, which are 8.1 times higher than the World Health Organization (WHO) recommendation for air quality every year (IQAir 2023). An example is the $\sim 170\ \mu\text{g}/\text{m}^3$ $\text{PM}_{2.5}$ observed at Baridhara (i.e., 34 times higher than WHO), east of Dhaka city, during the winter (Sarwar et al. 2023). Overall, the annual average $\text{PM}_{2.5}$ concentrations at this location are about five to six times higher than the Bangladesh standard ($15\ \mu\text{g}/\text{m}^3$), which is about eight to nine times higher than the WHO guideline ($10\ \mu\text{g}/\text{m}^3$). Chittagong, the second largest and commercial capital of Bangladesh, also reported crucial health impacts from air quality, especially from $\text{PM}_{2.5}$, which is highest ($321.1\ \mu\text{g}/\text{m}^3$) in the winter, during January, and more than five times higher than the WHO standard (Hossen and Hoque 2018). In another port city, Khulna, $\text{PM}_{2.5}$ concentrations are about $302\ \mu\text{g}/\text{m}^3$, which is also higher than the WHO air quality standard (Saju et al. 2023). This indicates that in different cities in Bangladesh, the concentration of $\text{PM}_{2.5}$ is much higher than the WHO standard. Moreover, the adverse health effects of prolonged exposure to $\text{PM}_{2.5}$ are well-documented and include respiratory diseases, cardiovascular problems, and even premature deaths, particularly affecting vulnerable populations such as children, the elderly, and individuals with pre-existing health conditions (Dibya et al. 2023; Xing et al. 2016).

Recognizing the urgency of the high $\text{PM}_{2.5}$ pollution levels in Bangladesh, the government, along with various international organizations and local non-governmental organizations (NGOs), has been actively monitoring and addressing $\text{PM}_{2.5}$ levels (World Bank 2022). Efforts have focused on implementing stricter emission standards, promoting cleaner technologies, enhancing public awareness, and developing sustainable urban planning strategies to mitigate air pollution through some projects (DOE 2019). In addition, Bangladesh's National Action Plan for Reducing Short-Lived Climate Pollutants, endorsed in 2018, outlines 11 priority measures across key sectors such as transportation and household energy to reduce black carbon

emissions by 72% by 2040, including actions like introducing clean biomass stoves and modernizing brick kilns (CCAC, 2018). Apart from these, Hoque et al. (2025) investigated the spatial distribution and diurnal variations of air pollutants, including particulate matter (PM_{2.5} and PM₁₀) and gaseous pollutants (CO, CO₂, CH₄, NO₂, and SO₂) over Dhaka, Bangladesh. Additionally, Zaman et al. (2021, 2025) examined the meteorological influences on PM_{2.5} concentrations, indoor air quality, and black carbon (BC) during winter in Dhaka. In this direction, up-to-date knowledge on PM_{2.5} air pollution, including its sources throughout the country, is required. Furthermore, PM_{2.5} pollution in Bangladesh poses severe health risks, including respiratory and cardiovascular diseases, increased mortality, cancer, neurological disorders, and preterm birth. Therefore, addressing these health impacts in Bangladesh is crucial and urgent. Efforts to reduce emissions, promote clean technologies, improve air quality monitoring, and raise public awareness are essential steps toward mitigating the health risks associated with PM_{2.5} pollution and improving the overall well-being of the population. Additionally, studies on PM_{2.5} in Bangladesh are limited and primarily focused on only a few cities. To this end, proper initiatives and thorough knowledge of PM_{2.5} air pollution across the country are needed. To achieve this, we study PM_{2.5} in Bangladesh and associated health impacts. Long-term spatiotemporal distributions, variations, and trends over 64 cities are investigated, and sources are identified. The study's novelty is its assessment of long-term PM_{2.5} trends alongside associated health risks, while incorporating new assessments of PM_{2.5} components and their links to meteorological conditions, industrial emissions, vehicle exhaust, and biomass burning. By comprehensively understanding PM_{2.5}-related issues, stakeholders can collaborate more effectively to combat this pressing environmental and public health challenge in Bangladesh (Goel et al. 2021).

Materials and methods

Study area

Bangladesh, situated in the tropical region of South Asia, covers an area from 88°0' to 92°42' E and 20°34' to 26°38' N (Fig. 1). India borders the country to the west, north, and northeast, Myanmar to the southeast, and the Bay of Bengal to the south. Bangladesh is divided into eight major administrative divisions (e.g., Dhaka, Rajshahi, Chattogram, Sylhet, Khulna, Barisal, Mymensingh, and Rangpur), which include 64 major cities and have a total population of about 174.76 million (Fig. 1a). The Chittagong Hill Tracts in Chittagong, the Sylhet Hills in Sylhet, and the hilly terrain of

Panchagarh in Rangpur are indicated in the elevation map in Fig. 1b. The country experiences tropical warm temperatures and thunderstorms during the pre-monsoon period (March–May), heavy rainfall during the South Asian Monsoon in the summer (June–August), and dry weather in the winter (December–February). Bangladesh is extremely susceptible to the impacts of global climate change (World Bank 2021). The rapid increase in population and economic activities has led to soil and air pollution becoming prominent emerging hazards. Bangladesh ranks among the world's most polluted countries, notably identified as an air pollution hotspot in the Indo-Gangetic Plain due to its dense population. Major contributors include industrial and vehicle exhaust emissions (Emberson et al. 2009; Mahmood 2011). Rapid industrial growth in Bangladesh, driven primarily by the textile and garment sector, has significantly contributed to the country's gross domestic product (GDP) but also worsened air pollution, with industrial activities accounting for approximately 30–40% of particulate matter (PM) pollution in major urban areas, notably impacting air quality and public health in cities like Dhaka (World Bank 2021). In addition to industrial pollution, brick kilns are a significant source of air pollution, particularly during dry seasons, resulting in elevated PM levels and other pollutants (Ali et al. 2022). Qiu et al. (2021) reported that air pollution in Bangladesh is largely attributed to emissions and the transport of pollutants from industrial cities across its administrative divisions, including Dhaka, Narayanganj, Gazipur, Rajshahi, Chattogram, and Khulna.

Datasets

In this study, Global Estimates of monthly PM_{2.5} using Geographically Weighted Regression (GWR) are used, developed by the atmospheric composition analysis team at Washington University (WU) (van Donkelaar et al. 2021). This dataset is produced by integrating satellite-based aerosol optical depth (AOD) estimates from the Moderate Resolution Imaging Spectroradiometer (MODIS), the Multiangle Imaging SpectroRadiometer (MISR), the Sea-viewing Wide Field-of-view Sensor (SeaWiFS), and the Visible Infrared Imaging Radiometer Suite (VIIRS) sensors, with the GEOS-Chem chemical transport model and ground-based AERONET measurements to derive ground-based PM_{2.5} using the GWR method. The PM_{2.5} data (2001–2020) were obtained from the link: <https://sites.wustl.edu/acag/datasets/surface-pm2-5/> (last access: 20 October 2023).

The present study utilized the Emissions Database for Global Atmospheric Research (EDGAR) v8.1_AP 2001–2020 dataset, which provides annual emissions data for key air pollutants (SO₂, NO_x, CO, NMVOC, NH₃, BC, and OC) by country and sector. The dataset was obtained from the

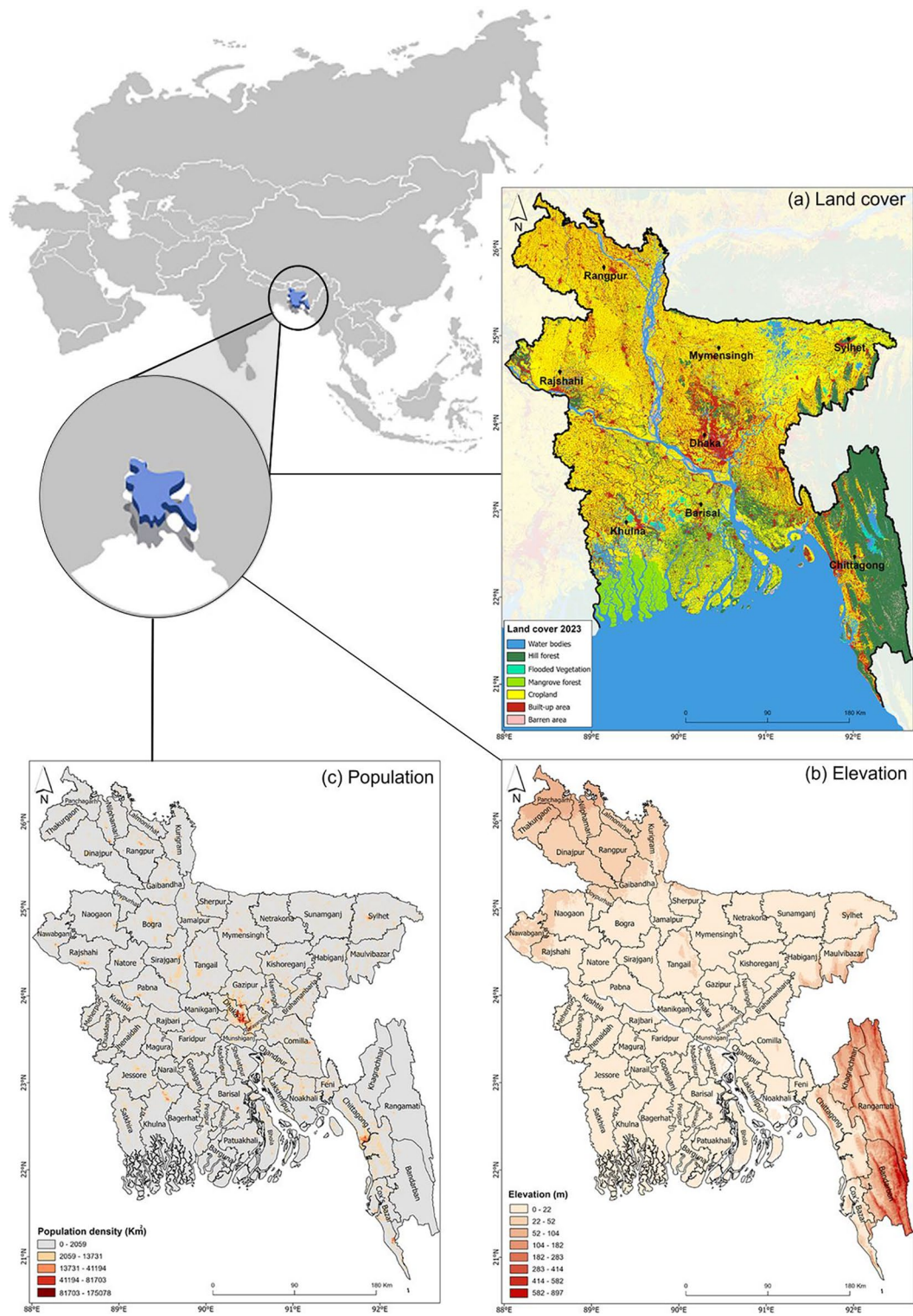


Fig. 1 Maps of Bangladesh showing **a**) Land cover, **b**) elevation (m), and **c**) population density (Km²). The locations of regions used in this study are identified by their names in these maps

European Commission (https://edgar.jrc.ec.europa.eu/emissions_data_and_maps) and was analyzed to assess its long-term interactions with PM_{2.5} concentrations. MERRA-2 reanalysis data (0.5° × 0.625° resolution) for ozone, temperature, wind speed, and surface mass concentrations of black carbon, dust, organic carbon, sulfate, and sea salt were obtained from NASA's GIOVANNI platform to analyze atmospheric composition and meteorological influences on PM_{2.5} levels. Monthly precipitation data at 0.5° resolution were also retrieved from the GPCP MON v3.2 satellite-gauge product, which integrates observations from AIRS, SSM/I, and ground-based measurements. These precipitation estimates, also accessed via GIOVANNI (<https://giovanni.gsfc.nasa.gov/giovanni/>), were analyzed to assess their long-term interactions with PM_{2.5} concentrations over Bangladesh. Relative humidity data were obtained from the NOAA Physical Sciences Laboratory (<https://psl.noaa.gov/data/atmoswrit/timeseries/>) to support the analysis of atmospheric conditions influencing PM_{2.5} concentrations.

Methodology

In the present study, the following methods were used:

- From 2001 to 2020, long-term annual and seasonal averages of PM_{2.5} were computed across the study area. Additionally, for both annual and seasonal analyses, the area-averaged PM_{2.5} for 64 cities in Bangladesh was derived using the city-level shapefile.
- A statistical quartile technique was used to characterize and rank 64 cities in Bangladesh by pollution levels, categorizing them as polluted (Q1) and extremely polluted (Q3). Cities with PM_{2.5} concentrations between Q1 and Q3 were classified as highly polluted cities (Bilal et al. 2021; Ali et al. 2023, 2024). For example, PM_{2.5} ≤ 59.64 (Q1) corresponds to polluted, 59.6 (Q1) < PM_{2.5} < 75.02 (Q2) represents highly polluted, and PM_{2.5} ≥ 75.02 (Q3) indicates extremely polluted.
- To determine trends and their significance, Theil-Sen's Slope (Sen 1968; Theil 1992) and bootstrapping methods were applied in conjunction with the Mann-Kendall (MK) test (Kendall 1975; Mann 1945), using a two-tailed test at a 95% confidence interval. Further information on the methodologies related to trend calculations can be found in Wang et al. (2021a, b).
- The study also calculated Health Risk (HR) assessment using the HR model (Kousa et al. 2002) by factoring in the population and population exposure to PM_{2.5}. The following equation was used to calculate HR:

$$P_{ek} = \sum_{f=1}^r M_{fk} D_{fk} \quad (1)$$

Here, P_{ek} indicates the total exposure of population k to a specific pollutant across various micro-environments f , M_{fk} represents the pollutant concentration ($\mu\text{g}/\text{m}^3$), r defines the number of different environments, and D denotes population density. Note that Eq. (1) implies that P_{ek} is the sum of the products of population density and PM_{2.5}. This signifies that as pollutant concentration and the number of exposed populations increase, the effective population grows, resulting in a higher death rate and increased HR (Peng et al. 2016; Tariq et al. 2023).

- To calculate the trend of changes in HR due to PM_{2.5} exposure, the study used unsupervised trend clustering (UnTC) analysis, a modified version of the traditional unsupervised classification (UC). UC is an automated technique that classifies data without needing training samples or prior knowledge of image features, requiring minimal user input (Redko et al. 2019). However, UnTC uses time-based data to segment images. It stacks images from different years, like layers, to study changes over time (Peng et al. 2016). The study utilizes the UnTC method, which categorizes the changing trend in health risks into four classes: slow rise (<2.4 μg . person/ m^3 /year), medium rise (2.4–2.5 μg . person/ m^3 /year), fast rise (2.5–2.6 μg . person/ m^3 /year), and rapid-fast rise (>2.6 μg . person/ m^3 /year). Further information on the UnTC method can be found in Tariq et al. (2023).
- Moreover, to manage air pollution, implementing PM_{2.5}-based air quality zones is essential (Peng et al. 2016; Mariam et al. 2021). Accordingly, four PM_{2.5} control zones were defined (Zones 1–4), which reflect the combined spatial extent of PM_{2.5} and population distribution. The WHO has defined air quality guidelines (AQG) and interim targets (IT) for outdoor PM_{2.5} (WHO 2021). Annual averages of PM_{2.5} categorize these standards with an AQG of 5 $\mu\text{g}/\text{m}^3$, and Interim Target-4 (IT-4) of 10 $\mu\text{g}/\text{m}^3$, Interim Target-3 (IT-3) of 15 $\mu\text{g}/\text{m}^3$, Interim Target-2 (IT-2) of 25 $\mu\text{g}/\text{m}^3$, and Interim Target-1 (IT-1) of 35 $\mu\text{g}/\text{m}^3$ (WHO 2021). However, the minimum PM_{2.5} exceeded the WHO guidelines during the study period in Bangladesh, and therefore, new twofold interim targets IT-2 for PM_{2.5} quantification are proposed as listed in Table 1.
- The HYSPLIT (Hybrid Single-Particle Lagrangian Integrated Trajectory) model (Stein et al. 2015) from

Table 1 PM_{2.5} control zones in Bangladesh

Zone	PM _{2.5} concentration ($\mu\text{g}/\text{m}^3$)	WHO guidelines
Zone 1	<50	<2IT-2
Zone 2	50–60	2IT-2–4IT-3
Zone 3	60–75	4IT-3–3IT-2
Zone 4	>75	>3IT-2

the National Oceanic and Atmospheric Administration (NOAA) (Fleming et al. 2012) was used to identify the sources of air masses. This model comprehensively simulates chemical transformation, dispersion, and transportation processes. The NOAA HYSPLIT model, integrated with a potential source contribution function (PSCF), was used to pinpoint potential sources of PM_{2.5} affecting air quality in Bangladesh. To this end, 72-hour HYSPLIT backward trajectories were calculated for every hour at 500 m above ground level (AGL) using meteorological data from the Global Data Assimilation System (GDAS) (link <http://arlftp.arlhq.noaa.gov/pub/archives/gdas1>; accessed: 23 July 2023) with a $1^\circ \times 1^\circ$ spatial resolution on a seasonal scale from 2001–2020. A height of 500 m is highly effective for representing pollution within the boundary layer (Begum et al. 2005). The 20-year PSCF analysis employed monthly PM_{2.5} on a 0.5° grid, and the PM_{2.5} threshold was calculated using the 75% percentile (e.g., $36.9 \mu\text{g}/\text{m}^3$). Further information on the PSCF method can be found in Bilal et al. (2021); Wang et al. 2021b); Ali et al. (2023).

Results and discussion

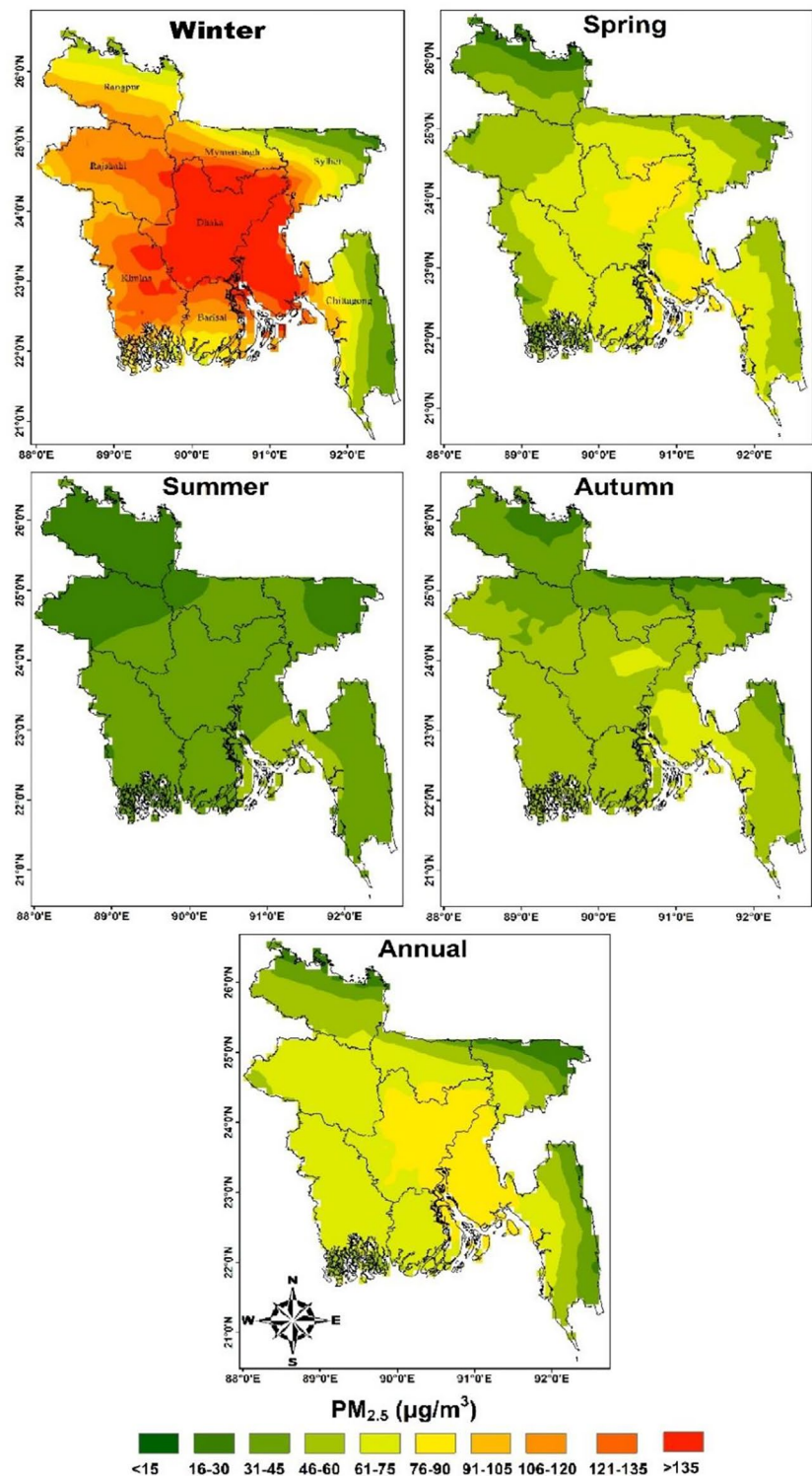
Spatial distribution of mean annual and seasonal PM_{2.5}

The spatial distributions of the annual and seasonal PM_{2.5} from GWR global estimates over Bangladesh, averaged over 20 years (2001–2020), are presented in Fig. 2. Overall, the spatial patterns of annually and seasonally averaged PM_{2.5} are similar, showing the highest concentrations in the urban areas of Dhaka, Mymensingh, Chittagong, Barisal, Khulna, Rangpur, and Rajshahi regions in Bangladesh. The elevated PM_{2.5} levels (annual mean $>46 \mu\text{g}/\text{m}^3$) in these regions are driven by urbanization, industrial growth, vehicular emissions, diesel generators, coal combustion, traditional brick kilns, construction dust, waste burning, and inadequate pollution controls (Raja et al. 2010; Arnab et al. 2013; Begum et al. 2018; Khatun et al. 2023; Jawaah et al. 2023). Begum et al. (2013) estimated that motor vehicles and brick kilns contribute 22% and 36%, respectively, to Dhaka's PM_{2.5}, with ~90% of PM_{2.5} mass stemming from human activities (Rahman et al. 2020, 2021). Apart from these, meteorological conditions such as temperature, humidity, wind speed, and boundary layer height significantly influence the spatial patterns of PM_{2.5} in Bangladesh, impacting their formation, dispersion, and accumulation (Islam et al. 2023). Several recent studies have reported that meteorological conditions significantly impact the spatiotemporal distributions of PM_{2.5} globally (Banks et al. 2022), in China (Cai et al. 2017;

Chen et al. 2019; Xu et al. 2020; Ali et al. 2023; Liu and Yang 2023), India (Jat and Gurjar 2021), Iran (Mohammadi et al. 2024), and Pakistan (Bilal et al. 2021). According to Wang et al. (2018), high PM_{2.5} concentrations are exacerbated by frequent air stagnation resulting from interactions between meteorological conditions and local terrain. However, spatially lower annual mean PM_{2.5} ($<31 \mu\text{g}/\text{m}^3$) were observed in the Chittagong Hill Tracts in Chittagong, the Sylhet Hills in Sylhet, and the hilly terrain of Panchagarh in Rangpur (Figs. 1b and 2), which is mainly due to lower population density, extensive vegetation cover, favorable meteorological conditions, limited industrial activity, and distance from major urban centers (European Environment Agency, 2018; Johnston et al. 2011; Chen et al. 2019; Ali et al. 2023). Apart from these observations, it is noteworthy that significant year-to-year variations in annual PM_{2.5} are prominent from 2001 to 2020, with higher PM_{2.5} concentrations in 2018 than in other years (Figure S1). A recent study conducted by Khatun et al. (2023) reported that the number of brick kilns in Bangladesh increased from 4500 fixed chimney kilns in 2009 to 8000 by 2018, while the number of motor vehicles increased by around 283% from 1.30 million in 2009 to 5.01 million in 2021.

The seasonal PM_{2.5} maps in Fig. 2 show much higher concentrations during the winter than in any other season. In winter, PM_{2.5} concentrations are often high ($>105 \mu\text{g}/\text{m}^3$) in many regions in Bangladesh due to temperature inversions and stagnant air, preventing ventilation of pollutants out of the area and thus causing accumulation of aerosols in a shallow boundary layer. Furthermore, biomass and agricultural residue burning, brick kilns activities and dust emissions increase during the winter while precipitation is reduced and thus the removal of aerosol particles by wet deposition (Begum et al. 2011, 2013; Guttikunda and Jawahar 2014; Begum and Hopke 2018; Ding et al. 2019). It is noteworthy that brick kilns contribute approximately 30–40% of the total PM_{2.5} emissions in Bangladesh during the winter months (Begum et al. 2011; Begum and Hopke 2019; Guttikunda & Khaliqzaman, 2013). In contrast, Guttikunda and Jawahar (2014) and Salam (2003) indicate that wintertime northwesterly winds can transport pollutants from India to Bangladesh, contributing to high PM_{2.5} in Bangladesh. A recent study by Khatun et al. (2023) reported that in winter, approximately 500 metric tons of dust settle on the ground daily, with around 2000 metric tons of dust remaining suspended in the air, significantly contributing to high PM_{2.5}. The springtime high PM_{2.5}, fluctuating between 46 and $90 \mu\text{g}/\text{m}^3$ in most regions of Bangladesh, is generally triggered by a combination of several factors, including seasonal agricultural practices, dust transportation from local sources and neighboring countries, and varying meteorological conditions (Guttikunda and Jawahar 2014). Similarly,

Fig. 2 Spatial distribution of annual and seasonal $PM_{2.5}$ ($\mu g/m^3$) over Bangladesh, averaged from 2001 to 2020



autumn high $PM_{2.5}$ varying between 46 and $75 \mu g/m^3$ in most regions of Bangladesh, except Rangpur, are influenced by post-monsoon conditions, agricultural residue burning, and changing meteorological patterns. Burning of the agricultural waste in the Indo-Gangetic Plain (IGP) from November to April (Autumn and Spring months)

contributes to high $PM_{2.5}$ in Bangladesh (Omni et al. 2017; Rahman et al. 2020). In addition, Begum et al. (2011) and Azkar et al. (2012) reported that brick kilns in Bangladesh operate in dry months from November to March, contributing to high $PM_{2.5}$ concentrations. Finally, abundant rainfall and high wind speeds in Bangladesh can lead to lower

PM_{2.5} in summer (Majumder et al. 2020; Wang et al. 2017a; 2017b). In addition to seasonal observations, significant month-to-month variations in PM_{2.5} are evident in most regions of Bangladesh, with the highest PM_{2.5} recorded in January, followed by February, December, and subsequent months (Figure S2). However, several factors, including temperature inversions, lower wind speeds, burning of agricultural residues, and increased emissions from brick kilns and vehicles, collectively may contribute to elevated PM_{2.5} in these months.

Annual and seasonal variations in city-level PM_{2.5}

The 20-year average annual and seasonal PM_{2.5} in 64 cities in Bangladesh are presented in Figs. 3. Figure 3 shows the differences between the 64 cities, which are ranked according to their annual mean concentrations. The annual averaged PM_{2.5} over the 64 cities varied between 2001 and 2020 from 43.55 to 84.10 $\mu\text{g}/\text{m}^3$, where 16 cities, Gazipur, Noakhali, Narshingdi, Dhaka, Narayanganj, Chandpur, Lakshmipur, Munshigonj, Comilla, Feni, Brahmanbaria, Kishoreganj, Shariatpur, Manikgonj, Bhola, and Tangail, are categorized as extremely polluted (Quartile, PM_{2.5} > 75.02). The annual averaged PM_{2.5} in the 64 cities is 9 to 17 times higher than the WHOAQs (annual mean: $\leq 5 \mu\text{g}/\text{m}^3$) and 3 to 6 times higher than the BNAAQS (annual mean: $\leq 15 \mu\text{g}/\text{m}^3$), indicating hazardous air quality for human health (Fig. 3). A recent study by AQLI (2023) reported that long-term exposure to PM_{2.5} reduces life expectancy, with citizens of Bangladesh losing more than 6.8 years on

average. Specifically, residents of Dhaka and Comilla lose more than 8 years on average, while those in Narayanganj lose 8.2 years, in Gazipur 8.8 years, and in Chittagong 6.6 years (Khatun et al. 2023; AQLI 2022).

On a seasonal basis, 61 cities in Bangladesh were categorized as extremely polluted in winter, 8 cities in spring, and none in summer or autumn (Fig. 3). In winter, the long-term (2001–2020) averaged PM_{2.5} ranged from 55.12 to 159.42 $\mu\text{g}/\text{m}^3$ across 64 cities in Bangladesh, with Gazipur having the highest value (categorized as an extremely polluted city) and Bandarban the lowest (polluted city). Moreover, PM_{2.5} in 64 cities is 11 to 32 times higher than the WHOAQs and 4 to 11 times higher than the BNAAQS, indicating severe hazardous air quality for human health (Fig. 3). In spring, PM_{2.5} ranged from 42.01 to 79.21 $\mu\text{g}/\text{m}^3$ across 64 cities in Bangladesh, with Gazipur having the highest value (extremely polluted city) and Lalmonirhat the lowest (polluted city). In this season, PM_{2.5} in 64 cities is 8 to 16 times higher than the WHOAQs and 3 to 5 times higher than the BNAAQS (Fig. 3). In summer, PM_{2.5} varied between 22.51 and 49.97 $\mu\text{g}/\text{m}^3$ across 64 cities in Bangladesh, with Noakhali having the highest value and Lalmonirhat the lowest, both classified as polluted (Fig. 3). The summertime PM_{2.5} in 64 cities is 4 to 10 times higher than the WHOAQs and 1 to 3 times higher than the BNAAQS. Finally, the autumn PM_{2.5} ranged between 34.08 and 65.21 $\mu\text{g}/\text{m}^3$ across 64 cities in Bangladesh, with Noakhali having the highest value (highly polluted) and Lalmonirhat the lowest (polluted), as shown in Fig. 3. The

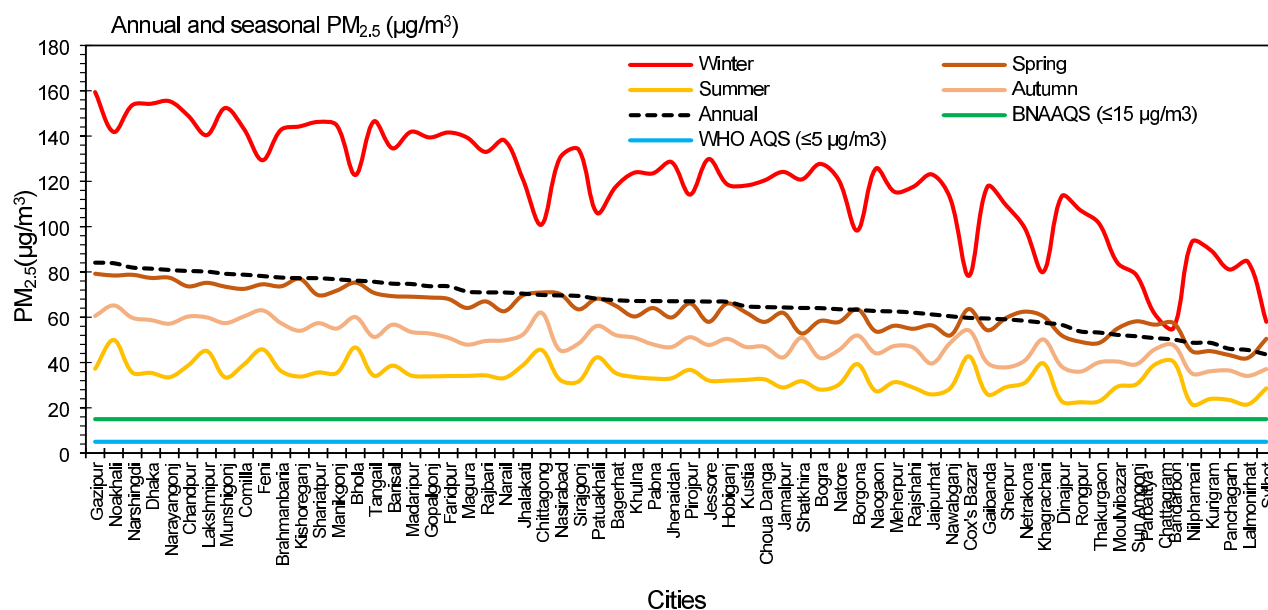


Fig. 3 Annual and seasonal PM_{2.5} averaged over the years 2001–2020, for 64 cities in Bangladesh, plotted along the horizontal axis and ranked according to PM_{2.5} concentration from high to low. The results

are compared with the BNAAQS (annual mean: $\leq 15 \mu\text{g}/\text{m}^3$) and WHO AQs (annual mean: $\leq 5 \mu\text{g}/\text{m}^3$)

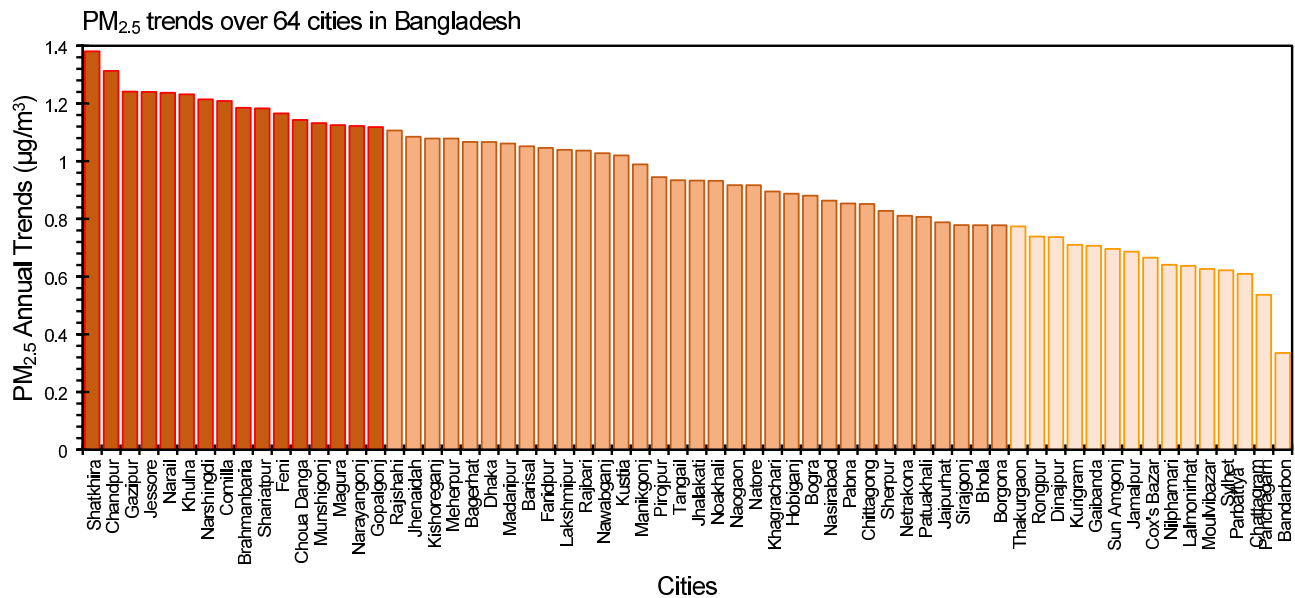


Fig. 4 PM_{2.5} annual trends (µg/m³) over 64 cities in Bangladesh from 2001 to 2020. All trends are statistically significant at a 99% confidence level

autumn PM_{2.5} in 64 cities is 7 to 13 times higher than the WHO AQS and 2 to 4 times higher than the BNAAQS.

Trends of PM_{2.5}

In this section, city-level PM_{2.5} trends are analyzed using Sen's Slope and the Mann-Kendall test to better understand the variability in PM_{2.5} across 64 cities in Bangladesh from 2001 to 2020, as presented in Fig. 4. During the entire analysis period, significant increasing trends in PM_{2.5} were observed in 63 cities in Bangladesh, apart from Bandarban. Specifically, a highly significant increase in PM_{2.5} was observed in several cities, with the highest rate in Shatkhira (1.38 µg.m⁻³/year), followed by Chandpur (1.31 µg.m⁻³/year), Gazipur, Jessore and Narail (1.24 µg.m⁻³/year), Khulna (1.23 µg.m⁻³/year), Narshingdi and Comilla (1.21 µg.m⁻³/year), Brahmanbaria and Shariatpur (1.18 µg.m⁻³/year), Feni (1.17 µg.m⁻³/year), Choua Danga (1.14 µg.m⁻³/year), Munshigonj and Magura (1.13 µg.m⁻³/year), Narayanganj and Gopalganj (1.12 µg.m⁻³/year), as shown in Fig. 4. Additionally, a moderately significant increase in PM_{2.5} was observed in several cities, with the highest rate recorded in Rajshahi (1.11 µg.m⁻³/year), followed by Jenaidah, Kishoreganj, and Meherpur (1.071 µg.m⁻³/year). In conjunction with these, a slightly significant increase in PM_{2.5} was observed in several cities, with the highest rate noted in Sirajgonj, Bhola, and Borgona (0.78 µg.m⁻³/year), followed by Thakurgaon (0.77 µg.m⁻³/year). It is important to note that PM_{2.5} in Bandarban increased (0.34 µg.m⁻³/year) but the trend is not statistically significant, likely due

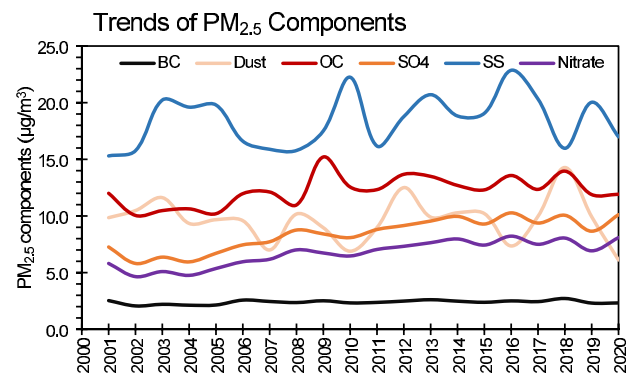


Fig. 5 Time sequences of MERRA-2 PM_{2.5} components in Bangladesh. BC: Black Carbon, OC: Organic carbon, SS: Sea Salt, SO₄: Sulfate, Nitrate: derived from sulfate

to its low population density, limited industrial activities, and extensive forest cover (World Bank 2022).

The time series of MERRA-2-based PM_{2.5} components in Bangladesh exhibit substantial interannual variability for Sea Salt (SS), Organic Carbon (OC), and Dust, whereas Sulfate (SO₄), Black Carbon (BC), and Nitrate show relatively smaller variations (Fig. 5). All components display an increasing trend, except for Dust, which shows an insignificant decreasing trend of -0.02 µg/m³/year. The highest rate of increase is observed for SO₄ at 0.23 µg/m³/year (significant at the 99% confidence level), while the lowest is seen for BC at 0.01 µg/m³/year, which is not statistically significant. Pavel et al. (2021) reported increasing trends in PM_{2.5}, PM₁₀, NO₂, and SO₂ concentrations over Dhaka, Bangladesh.

It is worth noting that numerous factors, such as weather conditions, industrial emissions, vehicle exhaust, and biomass burning, significantly influence PM_{2.5} concentrations, as illustrated in Fig. 6. In Bangladesh, climatic variables like temperature and relative humidity correlate positively with PM_{2.5}, whereas rainfall and wind speed show a negative correlation. A statistically significant negative correlation (at the 99% confidence level) between PM_{2.5} and rainfall highlights the scavenging effect of precipitation, which removes PM_{2.5} from the atmosphere. Most emissive aerosol pollutants are positively correlated with PM_{2.5}, except for carbon monoxide (CO) from transportation. Strong positive correlations ($r \approx 0.8$ or higher) are observed for CO-BB, BC-BB, OC-BB, NO_x-BB, SO₂-BB, and NO_x-transport, suggesting that biomass burning and regional transport are major contributors to these air pollutants.

PM_{2.5} health risk assessment and control zoning

Figures 7–9 shows spatial distributions of health risk (HR) due to PM_{2.5} exposure in 2001 and 2020, health risk changes between 2001 and 2020, and delineation of air quality control zones based on the WHOAQs for PM_{2.5} concentrations. However, instead of the 4 WHOAQ classes, we propose five HR classes, i.e. as very low risk ($<1.0 \times 10^4$ µg. person/m³), low risk ($1.0 \times 10^4 - 2.4 \times 10^4$ µg. person/m³), medium risk ($2.4 \times 10^4 - 4.9 \times 10^4$ µg. person/m³), high risk ($4.9 \times 10^4 - 7.4 \times 10^4$ µg. person/m³), and extremely-high risk ($7.4 \times 10^4 - 2325 \times 10^4$ µg. person/m³) as shown in Figs. 5a–b. The very low risk area, identified in the Chittagong Hill Tracts and Sundarban and Reserved Forest in Khulna, decreased from 13.90 to 10.95% between 2001 and 2020 (Fig. 7a–b; Table 2). Similarly, the low-risk area, primarily in Shylet, Mymensingh (Netrakona), Rangpur (Thakurgaon, Panchagarh), Rajshahi (Nawabganj), Khulna (the Sundarban and

Reserved Forest area), and Barisal (Borgona), and Chittagong (near the Hill Tracts), decreased from 13.98 to 5.41% between 2001 and 2020 (Fig. 7a–b; Table 2). Medium risk areas, which covered around 38.72% of Bangladesh in 2001, decreased to nearly 18.90% in 2020. The percentage of high-risk areas increased from 18.83% in 2001 to 25.45% in 2020. Similarly, the proportion of extremely high-risk areas in Bangladesh, including regions such as Dhaka, Mymensingh, Chittagong, Rangpur, Rajshahi, Khulna, and Barisal, increased significantly from 14.57 to 39.29% between 2001 and 2020 (Fig. 7a–b; Table 2). The significant increase in high and extremely high health risk areas in Bangladesh can be attributed to several important factors such as rapid urbanization, industrial activities, vehicles, brick kilns, deforestation, climate change, and population growth (Khatun et al. 2023; Rahman et al. 2020).

Additionally, health risks slowly increased between 2001 and 2020, by <2.4 µg.person/m³/year, which is observed in 31.31% of Bangladesh, predominantly in areas with low population density (around <100 – 500 people per km²) (Figs. 1a and 8, and Table 3). A medium health risk increase (2.4 – 2.5 µg.person/m³/year) was seen in 35.67% of Bangladesh, mainly in areas with a moderate population density (around 500 – 1500 people per km²) (Figs. 1a and 8). A fast rise rate (2.5 – 2.6 µg. person/m³/year) was identified in 19.94% of Bangladesh, primarily in regions with high population densities (around 1500 – 4000 people per km²) (Figs. 1a and 8, and Table 3). Finally, a rapid-fast rise rate (>2.6 µg.person/m³/year) was observed in 13.08% of Bangladesh, primarily in major urban areas with very high population densities (over 4000 people per km²).

Assessing air quality zones based on PM_{2.5} concentration is crucial for identifying pollution hotspots, mitigating PM_{2.5}-related pollution, and implementing targeted measures to protect human health (Peng et al. 2016; Mariam et

Fig. 6 Correlation between PM_{2.5} and meteorological element along with other emissive pollutants in Bangladesh. CO: Carbon Monoxide, BB: Biomass Burning, BC: Black Carbon, OC: Organic carbon, NO_x: Nitrogen Oxide, SO₂: Sulphur Di-oxide

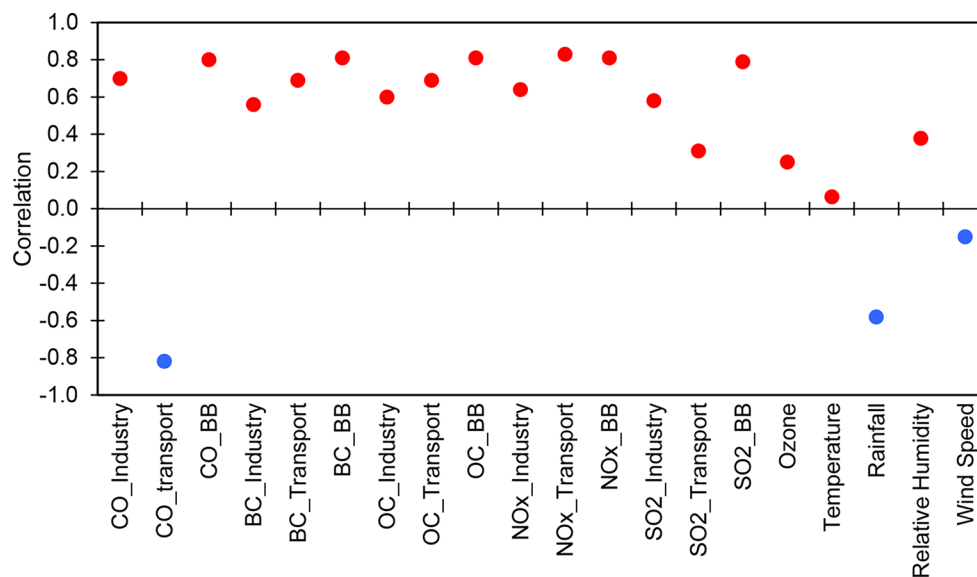
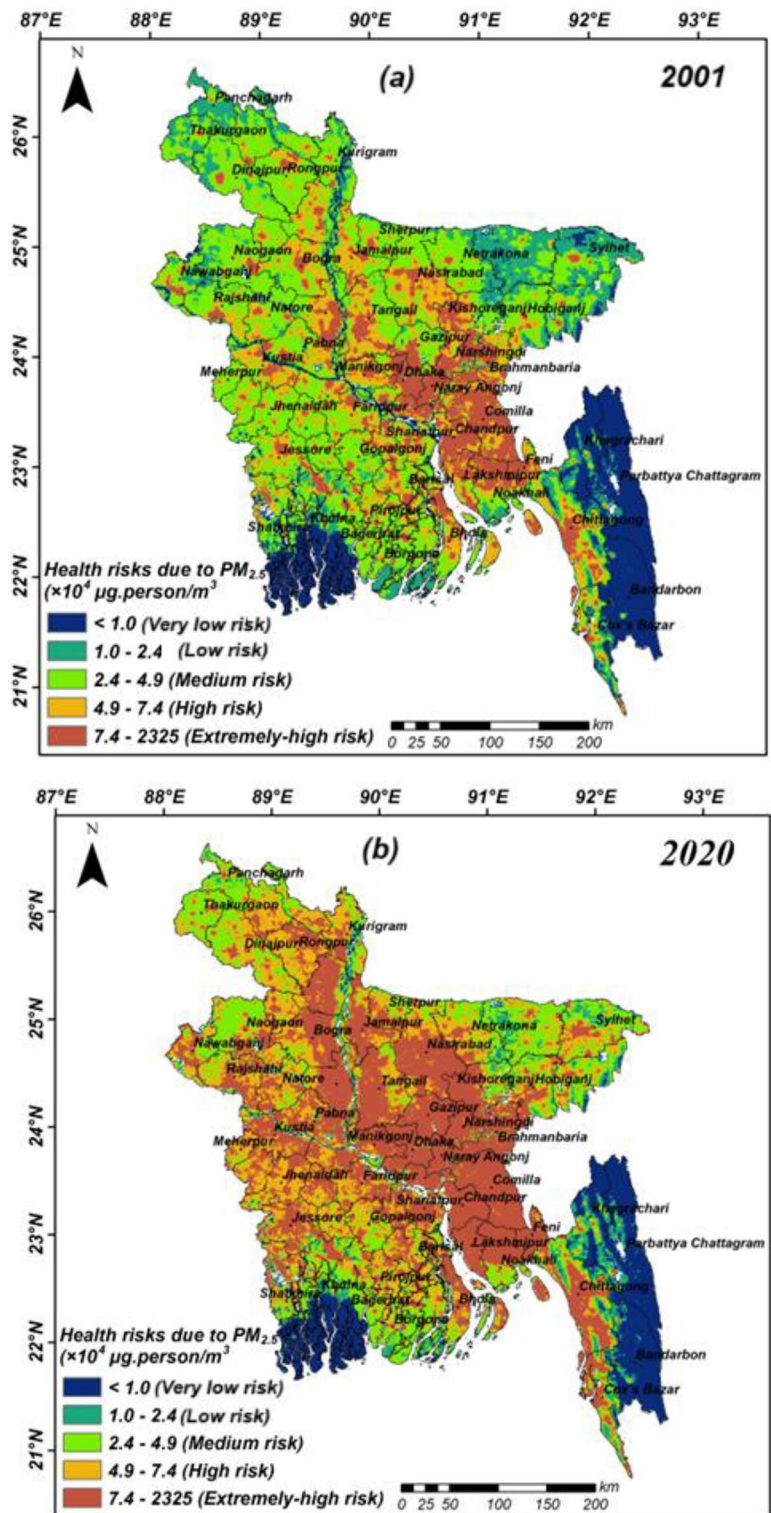


Fig. 7 Spatial distribution of **a)** health risks arising from $PM_{2.5}$ exposure in 2001 and **b)** in 2020 based on WHO air quality standards for $PM_{2.5}$ concentrations



al. 2021; Tariq et al. 2022). However, control zone 1, covering parts of Chittagong (the Chittagong Hill Tracts), Shylet, Mymensingh (Netrakona, Jamalpur, Sherpur), and Rangpur (Kurigram, Lalmonirhat, Nilphamari, Panchagarh), is exposed to $PM_{2.5}$ concentrations less than 2IT-2 ($< 50 \mu g/m^3$) (Fig. 9). Control zone 2, which includes regions such

as parts of Chittagong (Khagrachari, Parbattya Chattagram, Bandarban, Cox's Bazar), Shylet (Hobiganj), Mymensingh (Netrakona, Jamalpur, Sherpur), Rangpur (Dinajpur, Gaibandha, Rangpur, Thakurgaon), Rajshahi (Jaipurhat, Nawabganj), Khulna (Jessore, Shatkhira, the Sundarban and Reserved Forest area), and Barisal (Borgona), is exposed to

Table 2 Changes in area with different health risks due to PM_{2.5} exposure in Bangladesh between 2001 and 2020

Class	area (2001) in %	area (2020) in %	Difference (%)
Very Low risk	13.90	10.95	-2.95
Low risk	13.98	5.41	-8.57
Medium risk	38.72	18.90	-19.82
High risk	18.83	25.45	6.62
Extremely high risk	14.57	39.29	24.72

PM_{2.5} concentrations ranging from 2IT-2 to 4IT-3 (50–60 µg/m³) (Fig. 9). Control zone 3, the largest of the zones, covers regions such as Dhaka (Faridpur, Gopalganj, Tangail), Chittagong (Chittagong urban area), Mymensingh (Nasirabad, Mymensingh urban areas, Kishoreganj, Jamalpur), Rajshahi (Bogra, Naogaon, Rajshahi urban area, Natore, Pabna, Sirajgonj), Khulna (Meherpur, Kustia, Jhenaidah, Jessore, Khulna urban area), and Barisal (Bhola, Pirojpur, Barisal urban area). This zone is exposed to PM_{2.5} concentrations ranging from 4IT-3 to 3IT-2 (60–75 µg/m³) (Fig. 9). Control zone 4, primarily located in Dhaka (Dhaka urban areas, Gazipur, Manikgonj, Munshigonj, Narshingdi) and

Table 3 Changes in areas with different health risks between 2001 and 2020

Classes	%area	per year growth rate
Slow rise	31.31	<2.4
Medium rise	35.67	2.4–2.5
Rapid rise	19.94	2.5–2.6
Ultra-rapid rise	13.08	>2.7

Chittagong (Brahmanbaria, Comilla, Chandpur, Lakshmipur, Noakhali) regions, is exposed to PM_{2.5} concentrations greater than 3IT-2 (>75 µg/m³). In conclusion, these areas have encountered dangerously high levels of PM_{2.5} pollution, which poses significant health risks. To address this, it is vital to implement targeted strategies in Bangladesh to reduce PM_{2.5} concentrations, customized to the specific conditions of each region.

Bangladesh has introduced national strategies to reduce PM_{2.5}, including the 2018 National Action Plan for SLCPs and the recently approved National Air Quality Management Plan (2024–2030). These policies target key sectors like transport, household energy, waste, and agriculture by phasing out high-emission vehicles, promoting clean cookstoves, modernizing brick kilns, and improving waste

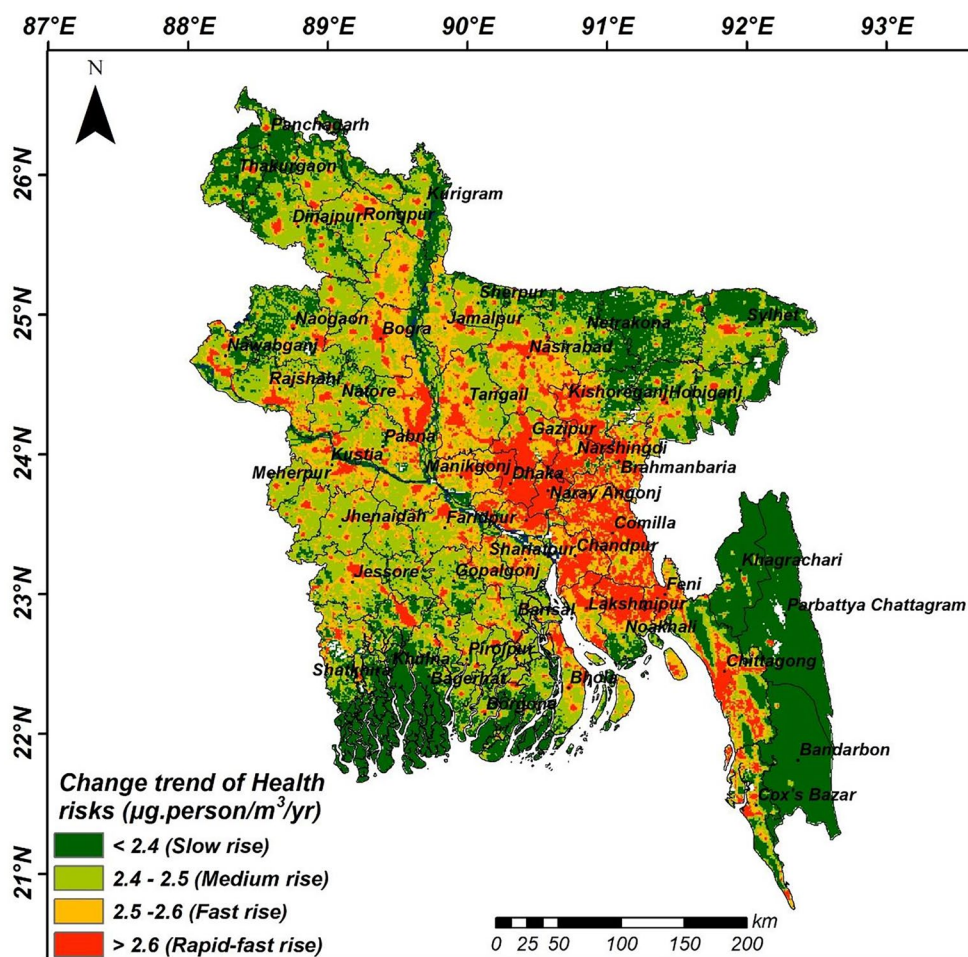
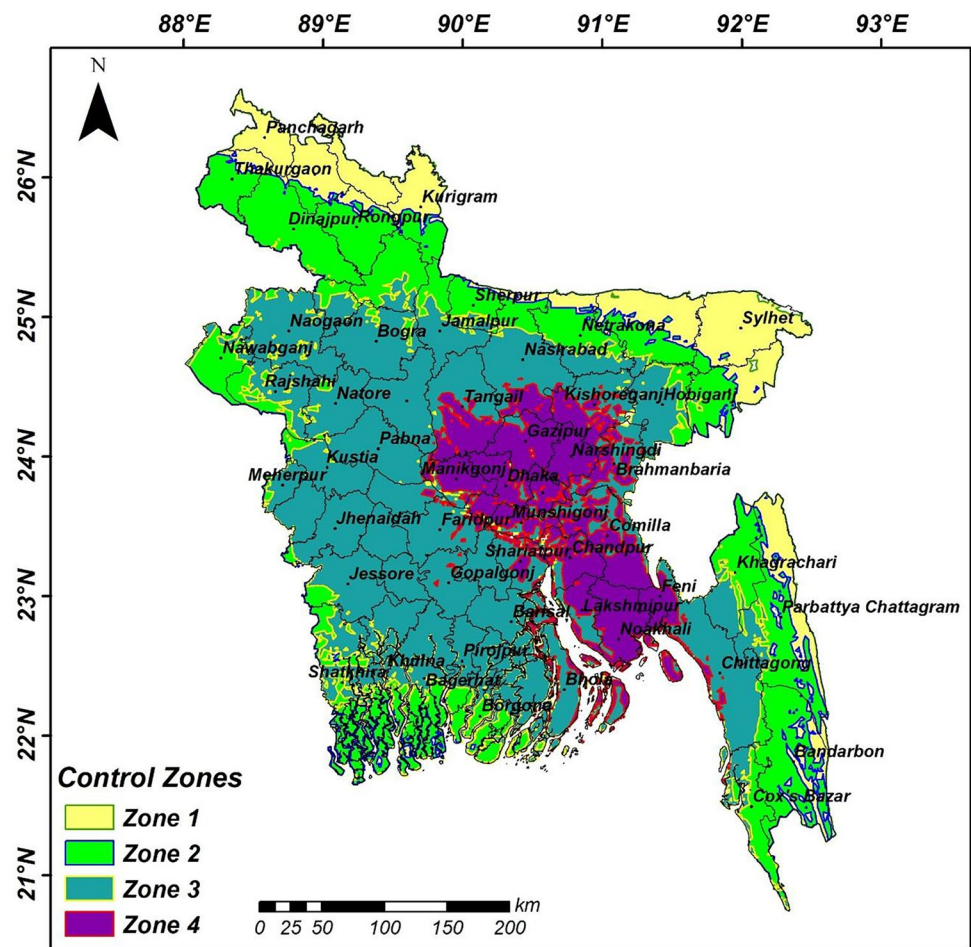
Fig. 8 Classification of the changing trend of health risks arising from exposure to PM pollution

Fig. 9 Spatial distribution of air quality control zones based on WHO air quality standards for $\text{PM}_{2.5}$ concentrations



management. However, enforcement, funding, and institutional capacity remain major challenges. In 2023, $\text{PM}_{2.5}$ levels averaged $79.9 \mu\text{g}/\text{m}^3$ —far above national ($35 \mu\text{g}/\text{m}^3$) and WHO ($5 \mu\text{g}/\text{m}^3$) standards—highlighting the gap between policy and impact. Furthermore, the absence of long-term, publicly available monitoring data limits the ability to assess effectiveness. Strengthened implementation and improved data infrastructure are essential for future progress.

Source identification of $\text{PM}_{2.5}$ using potential source contribution function (PSCF)

PSCF analysis using 72-hour back trajectories from the HYSPLIT model along with WU-based $\text{PM}_{2.5}$ was used to pinpoint potential $\text{PM}_{2.5}$ source regions for nine receptor sites: Dhaka, Rajshahi, Chittagong, Sylhet, Khulna, Barisal, Mymensingh, and Rangpur. The analyses are made for each season from 2001 to 2020, and the receptor site is indicated by the green arrows and stars (Fig. 6 for winter and Figs S3–S5 for the other seasons). Higher PSCF values suggest that more $\text{PM}_{2.5}$ is likely transported from that direction, implying stronger or more frequent source

contributions from those regions. The PSCF analysis shows significant variations across different areas and seasons in Bangladesh, with the air mass trajectories indicating long-range transport from the west (over approximately 1500 km or 15 degrees) for all nine locations, and for some locations, shorter-range transport from the southwest over ocean (over approximately 500 km or 5 deg). The long-range transport indicates that air quality in the eight major regions of Bangladesh is primarily influenced by external sources (Indo-Gangetic Plain or IGB, India, Nepal, Bay of Bengal, Sri Lanka, Gulf of Mannar, Arabian and Laccadive Sea) which are most pronounced in winter, followed by spring and autumn, and lowest in summer. External influences on air quality in Bangladesh are most pronounced in winter due to a combination of meteorological factors and regional pollution patterns. In winter, cold temperatures and stable atmospheric conditions, including temperature inversions, trap pollutants close to the ground, leading to higher concentrations of $\text{PM}_{2.5}$. Additionally, the winter monsoon winds transport pollutants to Bangladesh from the Indo-Gangetic Plain (IGP), where agricultural burning, industrial emissions, and vehicular pollution are prevalent. The impact

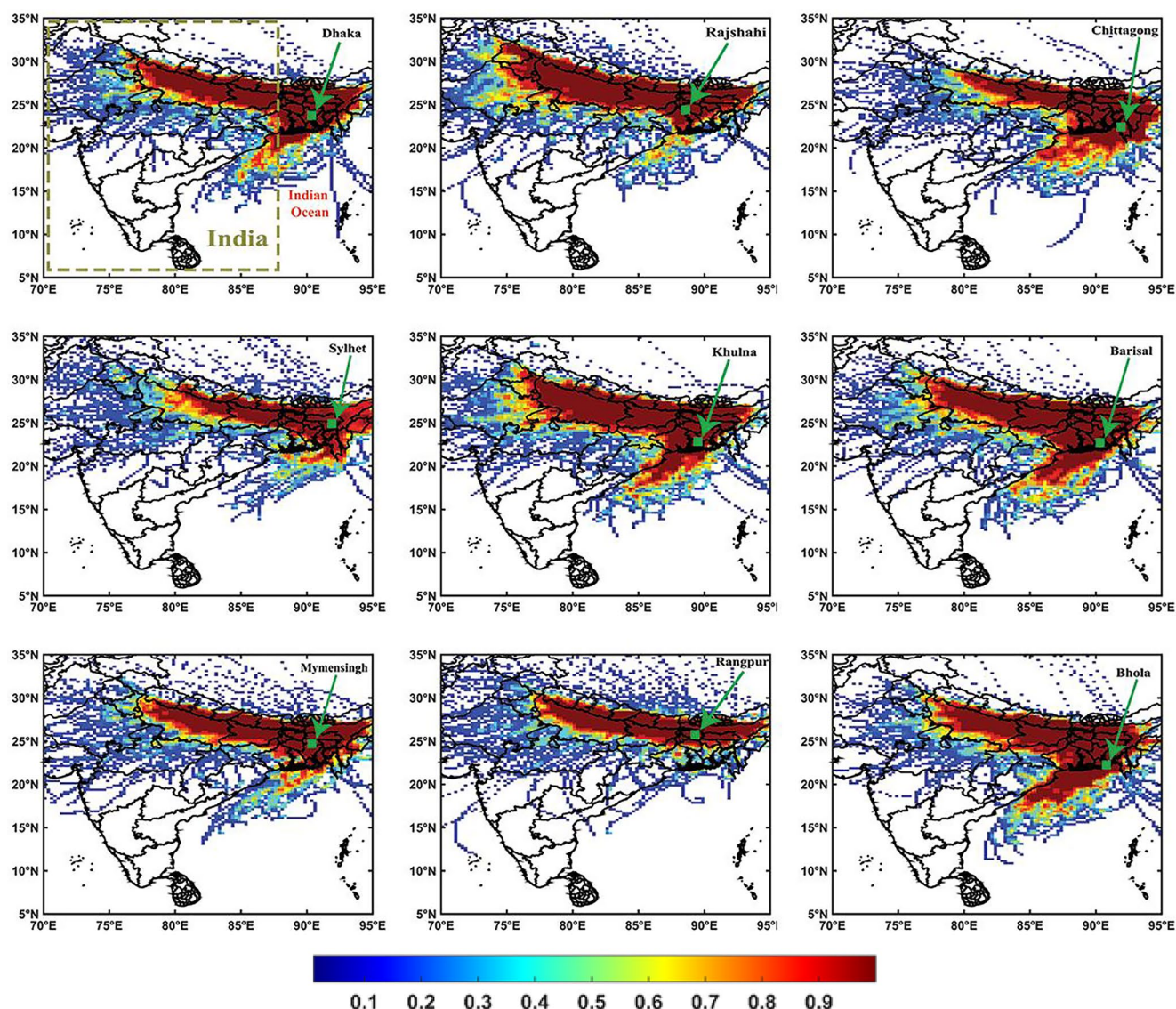


Fig. 10 Source identification of $PM_{2.5}$ over 9 major cities in Bangladesh using PSCF analysis during the winter season from 2001 to 2020. The arrows indicate the locations of the 9 receptor sites. The trajectory densities are color-coded according to the color bar at the bottom

is still significant in spring and autumn, but less intense as wind patterns shift and atmospheric mixing increases. Summer experiences the lowest external influences due to the monsoon rains, which remove pollutants by wet deposition, and strong winds disperse pollutants (Ali et al. 2022, 2023; Khatun et al. 2023) (Fig. 10).

Conclusions

This study aimed to examine the long-term (2001–2020) spatiotemporal distribution, variability, trends, health risk (HR), control zones, and potential source contribution function (PSCF) of $PM_{2.5}$ in Bangladesh, using GWR estimates of $PM_{2.5}$. The key findings of this study are as follows:

- Results from the long-term average spatial distribution analysis show the locations of $PM_{2.5}$ hotspots across the urban areas of Dhaka, Mymensingh, Chittagong, Barisal, Khulna, Rajshahi, and Rangpur in Bangladesh. The most frequent occurrences are in winter, followed by spring, autumn, and summer.
- City-level analysis shows that the annual mean $PM_{2.5}$ concentrations in 64 cities are 9 to 17 times larger than the WHOAQs and 3 to 6 times larger than the BNAAQS, indicative of hazardous air quality. Seasonally, 61 cities in Bangladesh were classified as extremely polluted ($PM_{2.5} \geq 75.02 \mu\text{g}/\text{m}^3$) in winter, 8 in spring, and none in summer or autumn. In winter, $PM_{2.5}$ concentrations in 64 cities across Bangladesh ranged from 55.12 to 159.42 $\mu\text{g}/\text{m}^3$, is 11 to 32 times higher than the

WHO AQs and 4 to 11 times higher than the BNA AQs. The springtime PM_{2.5} in 64 cities varied from 42.01 to 79.21 µg/m³, is 8 to 16 times higher than the WHO AQs and 3 to 5 times higher than the BNA AQs.

- Moreover, trend analysis shows significantly increasing concentrations in PM_{2.5} (0.54–1.38 µg/m³/year) between 2001 and 2020, in 63 cities in Bangladesh, with the highest rates observed in Shatkhira, followed by Chandpur, Gazipur, Jessore, Narail, Khulna, Narshingdi, Comilla, Brahmanbaria, Shariatpur, Feni, Choua Danga, Munshigonj, Magura, Narayanganj, and Gopalganj.
- PM_{2.5} components (BC, OC, Nitrate, SS, and SO₄) in Bangladesh show significant interannual variability, with all exhibiting increasing trends except for Dust.
- Weather conditions, industrial emissions, vehicle exhaust, and biomass burning significantly influence PM_{2.5} concentrations. For instance, most aerosol pollutants (particularly CO-BB, BC-BB, OC-BB, NO_x-BB, SO₂-BB, and NO_x-transport) exhibit strong positive correlations ($r \geq 0.8$) with PM_{2.5}, underscoring biomass burning and regional transport as major contributors.
- The health risk (HR) analysis shows that the percentage of areas with high-risk in Bangladesh increased from 18.83% in 2001 to 25.45% in 2020, while the areas with extremely high-risk, including Dhaka, Mymensingh, Chittagong, Rangpur, Rajshahi, Khulna, and Barisal, increased from 14.57% in 2001 to 39.29% in 2020.
- The air quality in Bangladesh is mainly impacted by external PM_{2.5} sources from IGB, India, Nepal, the Bay of Bengal, and surrounding seas, with the strongest effects in winter, followed by spring, autumn, and summer.

This study can be regarded as a base study to drive effective PM_{2.5} reduction in Bangladesh. Additionally, the country's authorities should prioritize strengthening air quality monitoring, enforcing emission controls, promoting clean energy, and enhancing regional cooperation to reduce PM_{2.5} effectively.

This study only focused on the physical transport of PM_{2.5}, ignoring chemical transformations during transport, such as secondary formation, aging, or interactions with co-pollutants, which can alter its composition, size, and health impacts. Future studies integrating in-situ chemical observations or chemistry-enabled modeling frameworks are recommended to capture these dynamics and provide a more comprehensive assessment of PM_{2.5}'s environmental and health effects. Future efforts should also incorporate observational datasets as they become more accessible to improve evaluation and model validation.

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Author Contribution AA prepared the plan of the paper. MA validated the results. NI and MB analyzed and interpreted the results. ST, YW, LA, AG, GL, and SS reviewed the manuscript. All authors read and approved of the final manuscript.

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Data availability Data will be available on request.

Declarations

Ethics approval Not applicable.

Consent to participate All authors consent to participate.

Consent for publication All authors consent to publish.

Competing interests The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

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